Sem IV	OE	Open Elective 1 (From Other School) Open-Source Technologies Introduction to Data Science	3	0	0	3	3
	PC	Design Thinking	0	0	2	2	1
	PC	Probability and Statistics Theory	3	0	0	3	3
	PC	Computer Organization and Architecture	3	1	0	4	4
	PC	Software Engineering	3	0	0	3	3
	PC	Software Engineering Laboratory	0	0	2	2	1
	PC	Theory of Computation	3	1	0	4	4
	PC	Design and Analysis of Algorithm	3	0	0	3	3
	PC	Design and Analysis of Algorithm Laboratory	0	0	2	2	1
			18	**2**	**6**	**24**	**23**
Sem V	OE	Open Elective 2 ((NPTEL/SWAYAM/MOOC))	3	0	0	3	3
	ESC	Introduction to Artificial Intelligence (Department Specific)	3	0	0	3	3
	HSC	Engineering Economics	3	0	0	3	3
	PE	Program Elective 1	3	0	0	3	3
	PC	Computer Networks	3	0	0	3	3
	PC	Computer Networks Laboratory	0	0	2	2	1
	PC	Compiler Design	3	0	0	3	3
	PC	Compiler Design Laboratory	0	0	2	2	1
	PC	Operating System	3	0	0	3	3
	PC	Operating System Laboratory	0	0	2	2	1
			21	**0**	**6**	**27**	**24**
Sem VI	OE	Open Elective 3 (From Other Department of FoET) Web Development Object Oriented Programming & Design	3	0	0	3	3
	PE	Program Elective 2	3	0	0	3	3
	PE	Program Elective 3	3	0	0	3	3
	PC	Cryptography and Network Security	3	0	0	3	3
	PC	Cryptography and Network Security Laboratory	0	0	2	2	1
	PC	Web & Mobile Development Essentials	1	0	0	1	1
	PC	Web & Mobile Development Essentials Laboratory	0	0	4	4	2
	PC	Distributed Computing	3	0	0	3	3
	PC	Distributed Computing Laboratory	0	0	2	2	1
			16	**0**	**8**	**24**	**20**
Sem VII	Pro	Summer Internship	0	0	0	0	2
	OE	Open Elective 4 (From Other Department of FoET) Green IT	3	0	0	3	3

		Global Data Security & Privacy Laws					
	ESC	Industry 4.0 (Department Specific)	2	0	0	2	2
	ESC	Industry 4.0 Laboratory (Department Specific)	0	0	2	2	1
	PE	Program Elective 4	3	0	0	3	3
	PE	Program Elective 5	3	0	0	3	3
	PC	Machine Learning	3	0	0	3	3
	PC	Machine Learning Laboratory	0	0	2	2	1
	PC	Cyber Laws and Ethics	1	0	0	1	1
	Pro	Seminar					1
			15	**0**	**4**	**19**	**20**
Sem VIII		**Course Name**	**Th**	**Tut**	**Pra**	**Hrs**	**Cr**
	Pro	Major/Comprehensive Project					**12**

Total Credits: 160

List of Electives (Track-based)

	Artificial Intelligence/ Machine Learning Track	**Industry Track**	**Speech, Vision & Text Track**	**Computing Track**	**Communication Track**
PE-1	Data Mining and Data Warehousing	Object Oriented Modelling and Design	Computer Graphics	Advanced Data Structure and Algorithms	Data Communication
PE-2	Soft Computing	UI/UX Design	Digital Image Processing	Blockchain Technology	Mobile Computing
PE-3	Deep Learning	Secure Software Engineering	Computer Vision	Big Data Analytics	Wireless Sensor Networks
PE-4	Agent based Learning	Agile and DevOps	Natural Language Processing	Cloud Computing	Internet of Things
PE-5	Machine Learning in Cyber Security	Web Application Testing	Speech Processing	Quantum Computing	Autonomous Systems

6th Semester

24CS321T					Web Development					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	-	-	100

COURSE OBJECTIVES

- To increase knowledge of available web technologies.
- To decorate the web pages using styling sheet.
- To become familiar with scripting languages.

UNIT 1: HTML & Forms Introduction To HTML, WWW, W3C, web publishing, Common HTML, Tags Physical & Logical, Some basic tags like <body>,changing background colour of page, text colour etc., Text formatting tags, <p> ,<hr> tags, Ordered & Unordered Lists Tags, Inserting image, Links: text, image links, image mapping, Tables, Frames, Form, Introduction with text box, buttons, radio buttons, Working with List box, checkbox, text area	**9 Hrs.**
UNIT 2: Cascaded Style Sheet Introduction to Style sheet, types of style sheets- Inline, External, Embedded CSS, text formatting properties, CSS Border, margin properties, Positioning, Use of classes in CSS, colour properties, use of & CSS background effects, padding properties, image properties	**10 Hrs.**
UNIT 3: Form Validation using Java Script Intro to script, types, intro of JavaScript, JavaScript identifiers, operators, control & Looping structure, Intro of Array, Array with methods, Math, String, Date Objects with methods, User defined & Predefined functions, DOM objects, Event handling, Validations On Forms	**10 Hrs.**
UNIT 4: Server-Side Scripting using PHP Introduction to PHP and the Web Server Architecture Model, HTML Embedding Tags and Syntax, Simple PHP Script, Decision Making, Flow Control and Loops, Working with Arrays Strings and functions, Outputting Data, Include and Require Statements, Built-in Database Functions, connecting to a MySQL, selecting a Database, Building and Sending the Query to Database Engine, Retrieving, Updating and Inserting Data	**13 Hrs.**
	42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Describe the basic html tags.
CO2 - Classify styling sheets as per their use.
CO3 - Apply CSS for web page design.
CO4 - Validate user input using Java Script.
CO5 - Develop server-side scripts using PHP
CO6 - Integrate Back-end with front-end using database connectivity.

TEXT/REFERENCE BOOKS

1. Thomas A. Powell, "Complete reference HTML", Tata McGraw-Hill Publications
2. Dave Mercer, Allan Kent, "Beginning PHP 5", Dreamtech Publication
3. David Flanagn, "JavaScript The definitive guide", SPD Publication

24CS322T					Object Oriented Programming & Design					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	-	-	100

COURSE OBJECTIVES

- To understand basics of Object-oriented programming.
- To Create classes for solving real-time problems.
- To implement code reusability using Inheritance.

UNIT 1: Basics OF Object-Oriented Programming Features of Object-Oriented Programming and Java, Basics of Java programming, Data types, Variables, Operators, Control structures including selection, Looping, Methods, Overloading, Math class, Arrays in java.	**10 Hrs.**
UNIT 2: Inheritance Basics of objects and classes in java, Constructors, Visibility modifiers, Inbuilt classes in Java, this reference; Inheritance in java, Overriding, Object class, Polymorphism, Dynamic binding, Abstract class, Interface in java, Package in java.	**10 Hrs.**
UNIT 3: I/O Programming, Exception Introduction to Java IO streams, Character and Binary streams, reading data from and writing data to files, Difference between error and exception, Exception handling in Java	**10 Hrs.**
UNIT 4: Event Handling and GUI Programming Event handling in Java, GUI Components, Flow Layouts, Grid Layout, Border Layout	**12 Hrs.**
	42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to
CO1 - Define the features of Object-oriented programming.
CO2 - Differentiate between Classes and Objects.
CO3 - Solve code redundancy issue using Inheritance.
CO4 - Distinguish between different types of Polymorphism.
CO5 - Summarize different exception handling mechanisms.
CO6 - Design a graphical user interface using Swing.

TEXT/REFERENCE BOOKS

1. Brett D. McLaughlin, “Headfirst Object-Oriented Analysis and Design”, O Reilly
2. Matt Weisfeld, “The Object-Oriented Thought Process”, Addison-Wesley Professional
3. Herbert Schild, “The Complete Reference, Java 2”, McGraw Hill
4. Balaguruswamy, “Programming with Java – A Primer”, McGraw Hill

24CS305T					Cryptography and Network Security					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	-	-	100

COURSE OBJECTIVES

- To learn fundamental concepts of security.
- To understand the mathematical concepts used to build cryptography application.
- To able to apply network security protocols to thwart the security vulnerability.

UNIT 1: Introduction Introduction to Cryptography, Modular Arithmetic, CIA, Classical Cryptography, Substitution and Transposition Techniques, Block Cipher and Stream Cipher, One time Pad, Pseudo Random Number generation.	**10 Hrs.**
UNIT 2: Modern Cryptography Polynomial Arithmetic, AES, Modes of Block Cipher Operations, Types of adversary models and attacks	**10 Hrs.**
UNIT 3: Public Key Cryptography System Mathematical Foundations for PKC; Euler's Totient Function, Fermat and Euler's theorem, Chinese Reminder Theorem, Euclidean and Extended Euclidean algorithm, RSA, Diffie-Hellman Key Exchange, MITM attack	**11 Hrs.**
UNIT 4: Digital Authentication and Network Security Cryptographic hash functions, Message authentication, Digital signature, Kerberos, X.509 authentication service, Email Security: S/MIME, PGP, IPSec, 802.11i.	**11 Hrs.**
	42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand CIA triod

CO2 - Use symmetric and asymmetric key algorithms for designing a security solution.

CO3 - Measure the security strength of a cryptography mechanism.

CO4 - Analyze Key Management techniques and importance of number Theory.

CO5 - Demonstrate the usage of the Message Authentication Codes and Hash Functions for digital authentication.

CO6 - Examine the issues related to network communication and propose solution for it.

TEXT/REFERENCE BOOKS

1. William Stallings, "Cryptography and Network Security Principles and Practice", Pearson Education.
2. Wade Trappe, Lawrence Washington, "Introduction to Cryptography with Coding Theory", Pearson Education
3. Bruce Schneier, "Applied Cryptography: Protocols, Algorithms, and Source Code in C", Wiley Computer Publishing.
4. Wenbo Mao, "Modern Cryptography: Theory and Practice", Prentice Hall.
5. Douglas Stinson, "Cryptography: Theory and Practice", Taylor & Francis.

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24CS305P					Cryptography and Network Security Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	-	-	-	50	50	100

COURSE OBJECTIVES

- To implement modular cryptography of security
- To understand the cryptographic building blocks.
- To apply the network security protocols to thwart the security attacks.

LIST OF EXPERIMENTS:

1. Download and Practice Cryptool.
2. Study and Implement program for Ceaser Cipher with Encryption, Decryption, Brute Force Attack, and Frequency Analysis functions.
3. Study and Implement a program for Transposition (Columnar) Cipher to encrypt and decrypt the message.
4. Study and Implement a program for Rail Fence Transposition Cipher to encrypt and decrypt the message.
5. Study and Implement a program for Vigenère Cipher to encrypt and decrypt the message.
6. Study and Implement a program for 6x6 Playfair Cipher.
7. Study and Implement a program for n-gram Hill Cipher.
8. Use Crypto++ library to implement encryption and decryption functions for different block ciphers.
9. Study and Implement RSA Encryption and Decryption function.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand the mapping of mathematical concepts with cryptography algorithms.

CO2 - Learn the modular cryptography building blocksDifferentiate the Symmetric and Asymmetric encryption techniques

CO3 - Evaluate the security strength of a cryptographic algorithm

CO4 - Compare the computational complexity of symmetric and asymmetric cryptography

CO5 - Implement Hashing techniques to create the message digest

CO6 - Utilize the network security tools to analyse network traffic

TEXT/REFERENCE BOOKS

1. William Stallings, "Cryptography and Network Security Principles and Practice", Pearson Education.
2. Wade Trappe, Lawrence Washington, “Introduction to Cryptography with Coding Theory”, Pearson Education
3. Bruce Schneier, “Applied Cryptography: Protocols, Algorithms, and Source Code in C”, Wiley Computer Publishing.
4. Wenbo Mao, “Modern Cryptography: Theory and Practice”, Prentice Hall.
5. Douglas Stinson, “Cryptography: Theory and Practice”, Taylor & Francis.

24CS306T					Web and Mobile Development Essentials					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
1	-	-	1	1	25	50	25	-	-	100

COURSE OBJECTIVES

- To design web and mobile applications.
- To develop web and mobile applications.
- To introduce Client-side scripting with JavaScript.

UNIT 1: Application Design Introduction to web technology, internet and www, Web site planning and design issues. HTML5: structure of html document, HTML elements: headings, paragraphs, line break, styles, colors, fonts, links, frames, lists, tables, images and forms.	**3 Hrs.**
UNIT 2: Data Validation JavaScript: Overview of JavaScript, Data types, Control Structures, Arrays, Functions and Scopes, HTML5 forms Validation, Objects in JS, DOM: DOM levels, DOM Objects and their properties and methods, Manipulating DOM.	**3 Hrs.**
UNIT 3: Server Programming and Database Connectivity Introduction to server side programming using PHP: Variable, Functions, Object oriented, Loops, Benefits of Database, Types of Database, steps to connect with Database	**4 Hrs.**
UNIT 4: Mobile App Development Android User Interface Design Essentials: User Interface Screen elements, Designing User Interfaces with Layouts, Drawing and Working with Animation, Testing Android applications, Publishing Android application, Using Android preferences, Managing Application resources in a hierarchy, working with different types of resource	**4 Hrs.**
	14 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Describe fundamental concepts of web technologies.
CO2 - Understand the validation mechanism and event handling using JavaScript.
CO3 - Analyse types of databases.
CO4 - Develop mobile application using Android.
CO5 - Use PHP and MySQL to integrate front end and backend.
CO6 - Design Android user interface by efficiently managing the resources.

TEXT/REFERENCE BOOKS

1. POWELL and THOMAS, “Html & Css: The Complete Reference”, McGraw Hill Publisher
2. Laurence Lars Svekis, Maaike van Putten, “JavaScript from Beginner to Professional: Learn JavaScript quickly by building fun, interactive, and dynamic web apps, games, and pages”, Packt Publishing
3. Greg Lim, “Beginning Android Development with Kotlin”, Greg Lim Publisher
4. Steven Holzner, “PHP: THE COMPLETE REFERENCE”, McGraw Hill Education
5. Wallace Jackson, “Learn Android App Development”, APress

24CS306P					Web and Mobile Development Essentials Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	4	2	4	--	--	--	50	50	100

COURSE OBJECTIVES

- To design web and mobile applications.
- To develop web and mobile applications.
- To introduce client side scripting with Javascript.

LIST OF EXPERIMENTS:

1 HTML page development
Create web page using HTML

2 Designing of Form
Create Form, Document using HTML

3 Form Validation in JS
Validation using java script

4 Implementation of AJAX
Development of AJAX script

5 Server-Side Scripting
Server-side programming Language using PHP, Object Oriented understanding using PHP

6 Database Connection
Implementation of database connection through php

7 Application Development
Mobile Application Development (Android Platform)-I
Mobile Application Development (IoS Platform)-I I

COURSE OUTCOMES

On completion of the course, student will be able to
CO1 - Describe the basic HTML tags.
CO2 - Validate the user input using JavaScript.
CO3 - Construct database for application development.
CO4 - Develop server-side response using PHP.
CO5 - Integrate frontend and backend using database connectivity.
CO6 - Develop Mobile Applications for different platforms.

TEXT/REFERENCE BOOKS

1. POWELL and THOMAS, “Html & Css: The Complete Reference”, McGraw Hill Publisher
2. By Laurence Lars Svekis, Maaike van Putten, “JavaScript from Beginner to Professional: Learn JavaScript quickly by building fun, interactive, and dynamic web apps, games, and pages”, Packt Publishing
3. Greg Lim, “Beginning Android Development with Kotlin”, Greg Lim Publisher
4. Steven Holzner, “PHP: THE COMPLETE REFERENCE”, McGraw Hill Education
5. Wallace Jackson, Learn Android App Development, APress
6. Jesse Feiler, Dummies , iOS App Development For Dummies, Willey

24CS307T					Distributed Computing					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25			100

PREREQUISITE: Operating Systems, Computer Network, Database Systems, Programming Skills

COURSE OBJECTIVES

- Students should gain an understanding of why distributed systems are important, their advantages, and challenges.
- To introduce principles, architectures and algorithms used in distributed systems
- To introduce how large-scale, distributed computational systems are built.
- The focus primarily is to examine state-of-the-art distributed systems.

UNIT 1 Introduction to Distributed Systems & Clocks Definition, Goals, Types, Architectures, Communication- Message-passing systems versus shared memory systems –Primitives for distributed communication –Synchronous versus asynchronous executions –Design issues and challenges, Distributed Computation. **Logical Clocks-**Scalar Time, Vector Time, Physical Clock Synchronization	**10 Hrs.**
UNIT 2 State Machines & Message Passing Global State and Snapshot Recording Algorithms, **Detecting Global Predicate**- Modalities on predicates, Centralized algorithm for relational predicates, Conjunctive predicates, Distributed algorithms for conjunctive predicates**; Election Algorithm**- Leader election in rings, Asynchronous leader election with identities, Synchronous leader election by abusing the synchronous model, **Message Ordering and Group Communication**- Termination Detection Algorithms and Reasoning with Knowledge	**10 Hrs.**
UNIT 3 Consistency, Consensus & Fault Tolerance **Distributed Mutual Exclusion Algorithms**- Lamport's algorithm, Ricart–Agrawala algorithm, and Deadlock Detection Algorithms, Distributed Deadlock Detection Algorithm, **Distributed Shared Memory**- Memory Consistency Models, Shared Memory Mutual Exclusion, Wait-Freedom Algorithm; **Byzantine Fault Tolerance, Checkpointing and Rollback Recovery**- Checkpoint-based recovery and Log-based rollback recovery; **Consensus and Agreement**- : Agreement in a failure-free system (synchronous or asynchronous) , Agreement in (message-passing) synchronous systems with Failures, Agreement in asynchronous message-passing systems with failures , Wait-free shared memory consensus in asynchronous systems;	**13 Hrs.**
UNIT 4 Distributed Storage & Case Studies **Distributed file servers**- GFS, MapReduce; **Selected case studies**- PNUTS, PAXOS, SPARK, Cryptocurrencies; **Advanced Topics**: ML in Distributed Systems-Tensorflow, Parameter Server	**9 Hrs.**
	Max. 42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1- Define the core concepts of distributed computing, including its goals, types, and architectural models.

CO2: Understand logical clock models such as scalar time, vector time and clock synchronization algorithms.

CO3: Design message passing algorithms for global state recording, snapshot recording, and detecting global predicates.

CO4: Evaluate distributed deadlock detection algorithms and their effectiveness in preventing system failures.

CO5: Analyse fault-tolerant distributed systems using Byzantine fault tolerance techniques, integrating checkpointing and rollback recovery mechanisms to safeguard system reliability and data integrity.

CO6: Evaluate case studies including PNUTS, PAXOS, SPARK, and cryptocurrencies, and advanced topics such as machine learning integration in distributed systems using frameworks like TensorFlow and Parameter Server.

TEXT/REFERENCE BOOKS

1. Ajay D Kshemkalyani and Mukesh Singhal, Distributed Computing: Principles, Algorithms, and Systems, Cambridge University Press.
2. Andrew S Tanenbaum, and Van Steen, Distributed Systems: Principles and Paradigms, Pearson Education India.
3. Sukumar Ghosh, Distributed Systems: An Algorithmic Approach, Chapman and Hall/CRC.

Additional Reference Material:

1. Abadi, et al (2016). TensorFlow: a system for large-scale machine learning. In Proceedings of the 12th USENIX conference on Operating Systems Design and Implementation (OSDI'16). USENIX Association, USA, 265–283..
2. Mu Li, et al. 2014. Scaling distributed machine learning with the parameter server. In Proceedings of the 11th USENIX conference on Operating Systems Design and Implementation (OSDI'14). USENIX Association, USA, 583–598.
3. Sanjay Ghemawat, Howard Gobioff, and Shun-Tak Leung. 2003. The Google file system. SIGOPS Oper. Syst. Rev. 37, 5 (December 2003), 29–43. https://doi.org/10.1145/1165389.945450
4. Gribble, Steven D. et al. "Scalable, distributed data structures for internet service construction." USENIX Symposium on Operating Systems Design and Implementation (2000).
5. Nakamoto, S. (2008) Bitcoin: A Peer-to-Peer Electronic Cash System.
6. Robbert van Renesse and Deniz Altinbuken. 2015. Paxos made moderately complex. ACM Comput. Surv. 47, 3, Article 42 et al.. 2008. PNUTS: Yahoo!'s hosted data serving platform. Proc. VLDB Endow. 1, 2 (August 2008), 1277–1288. https://doi.org/10.14778/1454159.1454167

24CS307P					Distributed Computing Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

PRE-REQUISITE:

- Proficiency in programming languages such as Java, Python, or C++.
- Familiarity with basic networking concepts.
- Proficiency in UNIX/Linux

COURSE OBJECTIVES

- Students should gain an understanding of how distributed systems are designed and developed.
- Provide students with practical exposure to various distributed system technologies, frameworks, and tools commonly used in industry settings.

TOOLS & TECHNOLOGIES

- Message Passing: MPI
- Distributed Storage and Processing: Hadoop MapReduce, HDFS
- Distributed Databases: Apache Cassandra, MongoDB
- Container Orchestration: Kubernetes
- Stream Processing: Apache Kafka, Apache Flink

1. Communication & Message Passing
- Implementing message passing using MPI (Message Passing Interface).
- Designing and implementing simple distributed algorithms for message passing.
- Assignment: Implementing parallel algorithms using MPI.

2. Implementing Byzantine fault tolerance algorithms.
- Designing and implementing consensus algorithms (e.g., Paxos, Raft).
- Assignment: Implementing fault-tolerant distributed applications.

3. Distributed Storage and Processing
- Implementing MapReduce algorithms for distributed data processing.
- Designing and implementing a distributed database application.
- Design and deploy a distributed application using container orchestration.

4. GROUP PROJECT

Max. 42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand the core concepts of distributed computing, major tools, technologies and frameworks.
CO2 - Apply Message Passing algorithm for synchronous and asynchronous communication.
CO3 - Apply distributed computing principles to design, develop, and implement distributed applications.
CO4 - Design and deploy distributed applications using virtualization or containerization tools..
CO5 - Evaluate the scalability, reliability, and efficiency of distributed applications under various workload conditions.
CO6 - Collaborate effectively within a team to design, implement, and test distributed system solutions.

TEXT/REFERENCE BOOKS

1. Nancy A. Lynch. Distributed Algorithms. Morgan Kaufmann Publishers Inc., San Francisco, CA, USA.
2. Andrew S Tanenbaum, and Van Steen, Distributed Systems: Principles and Paradigms, Pearson Education India.
3. Sukumar Ghosh, Distributed Systems: An Algorithmic Approach, Chapman and Hall/CRC.

Program Elective Course Track 1

Artificial Intelligence/ Machine Learning Track

TEXT/REFERENCE BOOKS

1. Tan, Pang-Ning, Michael Steinbach, and Vipin Kumar, "Introduction to data mining", Pearson Education.
2. Prabhu, " Data warehousing - concepts, Techniques, Products and Application s", Prentice Hall of India.
3. Alex Berson and Stephen J. Smith, "Data Warehousing, Data Mining & OLAP", Tata McGraw Hill
4. Gupta, Gopal K., "Introduction to data mining with case studies" PHI Learning Pvt. Ltd

Pandit Deendayal Energy University School of Technology

24CS336T					Soft Computing					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	-	-	100

COURSE OBJECTIVES

- To understand the concepts of feed forward &feedback neural networks
- To understand the concept of fuzziness involved in various systems
- To expose the ideas about genetic algorithm
- To provide adequate knowledge about of FLC and NN toolbox

UNIT 1 : Artificial Neural Network **Introduction:** Feed-forward, recurrent and multi-layer architectures, Supervised and unsupervised learning, **Characteristics:** Adaptability, fault tolerance, generalization, limitations of neuro-computing, **Perceptron:** Linear classifiers; Simple perceptron; Perceptron learning, algorithms; ADALINE; MADALINE; Limitation of perceptron dichotomizer.	**12 Hrs.**
UNIT 2: Multi-Layer Perceptron Network Multi-Layer Perceptron, Gradient decent scheme for error, minimization, Generalized delta learning rule, Back-propagation learning for multi-layer networks, Multi-layer perceptron's for multi- dimensional functional mappings, **Associated Memory Networks:** Auto-association; Hetero-association, **Linear associative networks:** Hebbian learning, perfect recall, cross-talk; Bidirectional associative memory, **Brain-State-in-a-Box network**, **Hopfield Networks:** Binary Hopfield network, basic structure, asynchronous updating, convergence, associative memory, Continuous-valued Hopfield network. Advantages and limitations, **Kohonen Networks:** Self-organizing networks, Similarity measures; Kohonen's winner-take-all network, Geometrical interpretation of Kohonen's learning, Functional specificity of human brain, Kohonen's self-organizing feature map algorithm; Conscience algorithm.	**12 Hrs.**
UNIT 3: Adaptive Resonance Theory ART and stability-plasticity dilemma; ART-1 architecture and algorithm, search, comparison and recognition phases, effect of vigilance, **Radial Basis Function Networks:** radial basis vs. linear basis, Gaussian basis functions, K-means learning, LMS algorithm, comparison with Multi-Layer Perceptron networks, **Support Vector Machines (SVM):** Optimal hyperplane for linear separability, quadratic optimization, SVM for pattern recognition, different kernels for hidden-layer, optimal design of SVM.	**09 Hrs.**
UNIT 4: Fuzzy Network Different faces of imprecision - inexactness, Ambiguity, Undecidability, Fuzziness and certainty, Fuzzy sets and crisp sets, Basic Definition, Fuzzy-set-theoretic Operations, Member Function, Formulation and Parameterization, Fuzzy Rules and Fuzzy Reasoning, Fuzzy propositions, Methods of decompositions and defuzzification, Fuzzy-neural networks, Neuro-fuzzy systems, Genetic algorithms: selection schemes, operations, hybrid algorithms,	**09 Hrs.**
	42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Ability to comprehend machine learning and soft computing techniques in solving real world applications
CO2 - Describe the differences between neural networks and fuzzy logic systems.
CO3 - Apply soft computing methods to solve real-world problems.
CO4 - Analyse the strengths and limitations of different soft computing approaches.
CO5 - Evaluate the performance of soft computing models using appropriate metrics.
CO6 - Synthesize knowledge from different soft computing techniques to propose hybrid systems.

TEXT/REFERENCE BOOKS

1. S.R.J ang, C.T. Sun and E. Mizutani, "Neuro-Fuzzy and Soft Computing", Prentice Hall of India and Pearson Education.

2. D.E. Goldberg, “Genetic Algorithms: Search, Optimization and Machine Learning”, Addison Wesley, New York.
3. S. Rajasekaran and G.A.V. Pai, “Neural Networks, Fuzzy Logic and Genetic Algorithms”, Prentice Hall of India.
4. R. Eberhart, P. Simpson and R. Dobbins, “Computational Intelligence - PC Tools”, AP Professional, Boston
5. Timothy J. Ross, “Fuzzy Logic with Engineering Applications”, Wiley India, 3rd edition, 2012.
6. S. Roy and U. Chakraborty, “Introduction to soft computing: neuro-fuzzy and genetic algorithms”, Pearson India

24CS341T					Deep learning					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- Understand the basic concepts of neural network.
- Understand the fundamental concepts, usage and impact of neural network, deep learning algorithms in various domains.
- Discuss various deep-learning algorithms to solve real life problems.

UNIT 1: Fundamentals of Neural Networks McCulloch Pitts Neuron, Multilayer Perceptrons (MLPs), Representation Power of MLPs, Sigmoid Neurons, Feed Forward Neural Networks, Back propagation, Gradient Descent (GD), Momentum Based GD, Nesterov Accelerated GD, Stochastic GD, Singular Value Decomposition, Parameters v/s Hyper-parameters, Learning Rule: Hebb’s Rule	**10 Hrs.**
UNIT 2: The Math behind Neural Networks Computation in neural network, The matrix magic, Programmatic expression of deep learning’s math constructs, Operations with the tensors, Array broadcasting, Scalar product/Inner product of tensors, Morphing shapes of tensors, Gradient calculation.	**10 Hrs.**
UNIT 3: Data Augmentation and Convolution Neural Network Data Augmentation, CNN, CNN Layers, Kernal, feature map, Pooling: Max Pool, Avg pool, Min Pool, Visualization of 2D convolution, Visualization of 3D convolution,	**10 Hrs.**
UNIT 4: Deep Learning Models Recurrent Neural Network, Long Short Term Memory, Gated Recurrent Unit, Boltzman Machine, Generative Adversarial Network	**12 Hrs.**
	42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to
CO1 - Learn how to modify state-of-the-art deep learning architectures for a new dataset/task.
CO2 - Know the basic model types used in deep learning, e.g., Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Generative Adversarial Networks (GANs).
CO3 - Understand the basic concepts of neural networks and deep learning methods.
CO4 - Know the suitability of specific deep learning methods to various real world data domains such as the ones arising from text, images, and videos.
CO5 - Re-train and tune hyperparameters of several classes of deep learning methods, in particular CNNs, RNNs, and GANs, on real-world datasets.
CO6 - Extract patterns from complex real world image and text datasets by using deep learning methods.

TEXT/REFERENCE BOOKS

1. Nikhil Buduma, “Fundamentals of Deep Learning: Designing Next-Generation Machine Intelligence Algorithm”, O’Reilly.
2. Ian Goodfellow, YoshuaBengio and Aaron Courville, “Deep Learning”, MIT Press.

Program Elective Course Track 2

Industry Track

24CS337T					UI/UX Design					
Teaching Scheme					**Examination Scheme**					
L	**T**	**P**	**C**	**Hrs/Week**	**Theory**			**Practical**		**Total Marks**
					MS	**ES**	**IA**	**LW**	**LE/Viva**	
3	**0**	**0**	**3**	**3**	**25**	**50**	**25**	**-**	**-**	**100**

PREREQUISITE: Web Technology; Software Engineering; Web Development and Mobile Application Development programming Skills

COURSE OBJECTIVES

- To understand User-Centered Design Principles.
- To gain a comprehensive understanding of UI design principles.
- To gain insights into designing for various platforms including web, mobile, and emerging technologies such as wearables and voice interfaces

UNIT 1 INTRODUCTION TO UI/UX DESIGN Understanding the basics of user interface and user experience design, Human-Computer Interaction (HCI) Principles	**10 Hrs.**
UNIT 2 USER RESEARCH AND PROTOTYPING Learning various user research techniques including interviews, surveys, and usability testing, Introduction to prototyping tools and techniques for creating wireframes and mockups.	**11 Hrs.**
UNIT 3 VISUAL DESIGN AND INTERACTION DESIGN Understanding color theory, typography, and layout in interface design, Designing intuitive and interactive user interfaces.	**11 Hrs.**
UNIT 4 ADVANCED TOPICS AND PROJECT MANAGEMENT Designing interfaces for various devices and screen sizes, Developing UX strategies and managing UI/UX projects effectively.	**10 Hrs.**
	42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand visual design principles and their application in UI/UX

CO2 - Comprehend HCI principles and their role in designing intuitive interfaces

CO3 - Demonstrate the ability to conduct user research using various methods such as interviews and surveys.

CO4 - Apply principles of typography, color theory, and layout to create visually appealing and cohesive interface designs.

CO5 - Develop skills in wireframing and prototyping

CO6 - Demonstrate proficiency in applying user interface (UI) and user experience (UX) design principles to create intuitive, user-friendly digital experiences

TEXT/REFERENCE BOOKS

1. Jesse James Garrett, The Elements of User Experience: User-Centered Design for the Web and Beyond, New Riders
2. Dan Saffer, Designing for Interaction: Creating Innovative Applications and Devices, New Riders
3. Saul Greenberg, Sketching User Experiences: The Workbook, Morgan Kaufmann
4. Robin Williams, The Non-Designer's Design Book, Peachpit Press Publications
5. Alan Cooper, About Face: The Essentials of Interaction Design, Wiley
6. Ethan Marcotte, Responsive Web Design, A Book Apart
7. Lean UX: Designing Great Products with Agile Teams by Jeff Gothelf and Josh Seiden.

24CS342T					Secure System Engineering					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	-	-	100

COURSE OBJECTIVES

- To learn fundamental concepts of secure systems
- To study various system vulnerabilities
- To study various software based vulnerabilities
- To study various application based vulnerabilities

UNIT 1: Introduction Introduction to gdb, buffer overflow, Preventing buffer overflow based malware, Return-to-libc attack, ROP Attacks	**10 Hrs.**
UNIT 2: Software based vulnerabilities Integer overflow and buffer overread and heap overflow, More on heap overflow; Access Control	**10 Hrs.**
UNIT 3: Application based vulnerabilities Confinement, SGX and Trustzone, Software fault isolation, Confinement in Applications, Mandatory access Control	**10 Hrs.**
UNIT 4: Hardware Security Micro-architectural Attacks, Hardware Security, Side Channel Analysis, Protecting against Hardware Trojans, Detecting Hardware Trojans in ICs	**12 Hrs.**
	42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand need of secure systems
CO2 - Explain hardware vulnerabilities models
CO3 - Apply mitigation techniques in systems
CO4 - Analyze security techniques in systems
CO5 - Analyze application vulnerabilities in systems.
CO6 - Determine attack analysis in systems

TEXT/REFERENCE BOOKS

1. "Introduction to Computer Security" by Matt Bishop, Pearson Publications.
2. Security Engineering: A Guide to Building Dependable Distributed Systems by Ross J. Anderson
3. NPTEL course : https://archive.nptel.ac.in/courses/106/106/106106199/

Program Elective Course Track 3

Speech, Vision & Text Track

24CS338T					Digital Image Processing					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To introduce mathematical modelling of transforms for digital image processing.
- To understand methodology to analyse, design and implement images processing and analysis algorithms.
- To develop understanding for image preprocessing methodologies like segmentation, noise elimination and edge detection.

UNIT 1: Fundamental And Spatial Domain Processing Introduction, Image sampling and quantization, Basic relationships in pixels, Basic intensity transformations, Histogram processing: local and global, Spatial filtering: smoothing and sharpening, Image Enhancement.	**10 Hrs.**
UNIT 2: Frequency Domain Processing Sampling 2D functions, 2D Fourier Transform and properties, Filtering in frequency domain: smoothing, sharpening. Performance comparison between band pass, gaussian and notch filters in both high and low pass filtering applications.	**10 Hrs.**
UNIT 3: Image Restoration And Morphological Operations Model of image degradation process, Restoration in presence of noise, estimating degradation function, Inverse filtering, MMSE filtering, Morphological operations: Erosion and dilation, opening and closing, Performance comparison between morphological and restoration techniques algorithms for noisy images.	**10 Hrs.**
UNIT 4: Image Segentation And Description Point, line and edge detection, Thresholding, Basic segmentation algorithms: region based, watershed, Image representation: chain codes, polygonal approximation, Boundary and regional descriptors: textures, moments, Feature extraction using shape features. Performance comparison between thresholding and region-based segmentation algorithms.	**12 Hrs.**
	42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand properties of an image and its digitization.
CO2 - Understand image processing in spatial and frequency domain.
CO3 - Apply image sharpening and smoothing for image enhancement.
CO4 - Apply spatial and frequency domain filtering for image denoising and restoration.
CO5 - Demonstrate morphological operations like opening and closing on images for connected component extraction in real world applications.
CO6 - Apply edge detection algorithm and thresholding for object segmentation for real world applications.

TEXT/REFERENCE BOOKS

1. Gonzalez, R. C., & Woods, R. E., “Digital image processing”, Pearson.
2. Sonka, Milan, “Image processing, analysis and machine vision”. Cengage Learning Pvt. Ltd.
3. Jayaraman, “Digital Image Processing”. McGrawhill.
4. Gose, Earl, “Pattern recognition and Image Analysis” PHI Learning Pvt. Ltd.
5. Alasdair, McAndrew, “A Computational Introduction to Digital Image Processing”. CRC Press.
6. Artyom M ,Grigoryan, “Image Processing”. Taylor & Francis Ltd.
7. Castleman, Kenneth “Digital Image Processing”. Pearson Education.

24CS343T					Computer Vision					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	-	-	100

PREREQUISITE: Linear Algebra

COURSE OBJECTIVES

- Ability to work with features above the pixel level.
- Develop ability to understand the difference in theory and practice of Computer Vision.

UNIT 1: Low Level Vision Introduction, Applications and Challenges, Fundamentals of Image Formation, Pin hole camera, Camera Parameters, Geometric camera calibration, Transformation: Orthogonal, Euclidean, Affine, Projective, Convolution and Filtering, Image Enhancement, Histogram Processing, Stereo Vision, Photometric Stereo.	**10 Hrs.**
UNIT 2: Mid Level Vision Edge and corner detection, Hough Transform, SIFT, SURF, HOG, Image Segmentation: Region Growing, Edge based methods, Graph-Cut, Mean-Shift, K-Means, K-Medoids, EM algorithm.	**10 Hrs.**
UNIT 3: High Level Vision Error, Loss, Image Classification, Object detection: Face Detection, State of the art in Object detection, Object Recognition, Hidden Markov Model, Object Tracking, Background Subtraction and Modeling, Optical Flow, KLT, Spatio-Temporal Analysis, Dynamic Stereo, Motion parameter estimation.	**10 Hrs.**
UNIT 4: Computer Vision Using Deep Learning Convolutional Neural Networks architectures, Convolution and Pooling, training CNN, Data Augmentation, Transfer Learning, Recurrent Neural Networks, LSTM, GRU, Applications like Image Captioning, Face Recognition, Semantic Segmentation.	**12 Hrs.**
	42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Define low level to high level vision.

CO2 - Describe basic methods of image formation and image processing including camera projections, image filtering, and feature extraction.

CO3 - Develop applications like classification, semantic segmentation, tracking, person identification.

CO4 - Choose appropriate computer vision method for a given problem statement.

CO5 - Apply computer vision techniques for solving practical problems.

CO6 - Implement vision models based on deep neural networks.

TEXT/REFERENCE BOOKS

1. Richard Szeliski, Computer Vision: Algorithms and Applications, Springer-Verlag London Limited 2011.
2. D. A. Forsyth, J. Ponce, Computer Vision: A Modern Approach, Pearson Education, 2003.
3. Simon Prince, Computer Vision: Models, Learning and Interface, Cambridge University Press.

Program Elective Course Track 4

Computing Track

24CS339T					Blockchain Technology					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	-	-	100

COURSE OBJECTIVES

- To understand the concepts of distributed consensus and trust management.
- To understand the design principles of the block chains.
- To design and implement the distributed ledger and the smart contracts.

UNIT 1: Blockchain Fundamentals Basic Data Structure concepts related to blockchain. Introduction to Blockchain, Cryptographic primitives used in Blockchain. Basic Distributed System concepts – distributed consensus and Byzantine fault-tolerant consensus methods.	**11 Hrs.**
UNIT 2: Blockchain Types And Cryptocurrency Proof-of-Work based consensus mechanisms, Proof of Stake based Chains, Types of Blockchain. Introduction to Crypto Currency, Crypto Currency as application of blockchain technology.	**11 Hrs.**
UNIT 3: Smart Contract And Blockchain Platform Blockchains with smart contracts - Ethereum platform and its smart contract mechanism.	**10 Hrs.**
UNIT 4: Blockchain Use Cases Blockchain Use Cases – Finance, Industry, E-Governance and other contract enforcement mechanisms. Security and Research Aspects in Blockchain.	**10 Hrs.**
	42 Hrs.

COURSE OUTCOMES

At the end of the course, students will be able to -

CO1 - Define the role of Blockchain technology in digitization.
CO2 - Illustrate the cryptographic concepts, and distributed concepts related to Blockchain technology.
CO3 - Experiment with Ethereum framework for Blockchain development.
CO4 - Analyze the need of Blockchain for real life system.
CO5 - Choose the appropriate type of Blockchain, and framework according to Blockchain usecase.
CO6 - Create the smart contracts and Blockchain for suitable system.

TEXT/REFERENCE BOOKS

1. Josh Thompson, 'Blockchain: The Blockchain for Beginnings, Guide to Blockchain Technology and Blockchain Programming', Create Space Independent Publishing Platform
2. Mark Gates, "Blockchain: Ultimate guide to understanding blockchain, bitcoin, cryptocurrencies, smart contracts and the future of money", WiseFox publishing.
3. Debajani Mohanty, "Ethereum for Architects and Developers: With Case Studies and Code Samples in Solidity", Apress.

24CS344T					Big Data Analytics					
Teaching Scheme					**Examination Scheme**					
L	**T**	**P**	**C**	**Hrs/Week**	**Theory**			**Practical**		**Total Marks**
					MS	**ES**	**IA**	**LW**	**LE/Viva**	
3	**0**	**0**	**3**	**3**	**25**	**50**	**25**	**-**	**-**	**100**

COURSE OBJECTIVES

- Learn the issues of Big Data.
- Learn to build and maintain reliable, scalable, distributed systems with Hadoop.
- Able to apply Hadoop ecosystem components.
- Working with SPARK for Data Analysis.

UNIT 1: Introduction Introduction to Big Data, different Vs in Big Data Analytics, Data Serialization, Apache Hadoop & Hadoop Ecosystem, Introduction to MAP-Reduce Architecture, Analysing Data with Hadoop, Hadoop Streaming	**11 Hrs.**
UNIT 2: Distributed File System Distributed File System, the Design of HDFS, HDFS Concepts, Interfaces, Data flow, Data Ingest with Flume and Scoop and Hadoop archives, Hadoop I/O, Avro and File-Based Data structures. Introduction to Scala programming and platforms.	**10 Hrs.**
UNIT 3: Processing Of Big Data Anatomy of a Map Reduce Job Run, Map Reduce Types and Formats, Map Reduce Features, Failures, Job Scheduling, Shuffle and Sort, Matrix-Vector multiplication by Map-reduce, Task Execution, Introduction to PIG, HIVE, Kafka.	**11 Hrs.**
UNIT 4: Data Analysis With Spark SPARK Introduction to Data Analysis with Spark, Downloading Spark and Getting Started, Programming with RDDs, Machine Learning with MLlib, Introduction to NoSQL tools, Data Visualization tools, Analytics using R programming.	**10 Hrs.**
	42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand the fundamental concepts of Big Data management and analytics
CO2 - Learn the concepts and methodologies for Map Reduce Programming
CO3 - Evaluate tools of Hadoop echo system
CO4 - Identify and exploit different features of SPARK for big data applications
CO5 - Apply the skills to conduct different types of data analysis
CO6 - Develop effective strategies for predictive and streaming data analysis for real world applications

TEXT/REFERENCE BOOKS

1. Chris Eaton, Dirk deroos et al., “Understanding Big Data”, McGraw Hill.
2. Tom White, “HADOOP: The definitive Guide”, O Reilly.
3. Boris lublinsky, Kevin t. Smith, Alexey Yakubovich, “Professional Hadoop Solutions”, Wiley.
4. Donald Miner & Adam Shook, “MapReduce Design Patterns”
5. Bill Chambers, Matei Zaharia, “Spark: The Definitive Guide”, O'Reilly Media, Inc

Program Elective Course Track 5

Communication Track

24CS340T					Mobile Computing					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	-	-	100

COURSE OBJECTIVES

- To understand the basic concepts of mobile computing.
- To study the various generation of the mobile communication system.
- To demonstrate the various cellular networking architecture.
- To illustrate the working of various routing protocols in mobile networks.

UNIT 1: Introduction Introduction to Mobile Computing, Applications of Mobile Computing, Generations of Mobile Communication Technologies and their technical specification, Multiplexing -MAC Protocols SDMA- TDMA- FDMA- CDMA	**10 Hrs.**
UNIT 2: Cellular Mobile Wireless System Introduction to Cellular Systems, GSM- Multiple Access Scheme, GSM Protocols and Signalling, Connection Establishment Frequency Allocation, Routing, Mobility Management, GPRS- Protocols, Network Architecture, UMTS- Network Protocol Architecture, Mobility Management.	**12 Hrs.**
UNIT 3: Mobile Ad-Hoc Network Mobile IP- Operations, Registration, Tunnelling, Proactive protocol- Destination-Sequenced Distance-Vector Routing, Reactive Routing Protocols- Dynamic Source Routing, Ad hoc On-Demand Distance Vector, Hybrid routing- Zone Routing Protocol, Multicast Routing- On-Demand Multicast Routing Protocol, Vehicular Ad Hoc Networks, MANET Vs VANET	**10 Hrs.**
UNIT 4: Mobile Transport and Application Layer Mobile TCP, Wireless Application Protocol, Wireless Transport Layer Security, Wireless Transaction Protocol, Wireless Application Environment, Wireless Telephony Application.	**10 Hrs.**
	42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand the fundamental concepts and principles of mobile computing.
CO2 - Illustrate the GSM, GPRS and UMTS cellular networking architecture.
CO3 - Discuss the mobility management in the cellular networks.
CO4 - Analyze the proactive and reactive routing strategies in the Ad-hoc network.
CO5 - Compare the Vehicular Ad-hoc Networks with Mobile Ad-hoc Networks.
CO6 - Determine the functionality of transport and application layer protocols.

TEXT/REFERENCE BOOKS

1. Jochen Schiller – Mobile Communication –PHI
2. Prasant Kumar Pattnaik, Rajib Mall- Fundamentals of Mobile Computing, PHI
3. A.Goldsmith, Wireless Communications, Cambridge University Press.
4. Siva Ram Murthy C and Manoj B S, “Ad Hoc Wireless Networks: Architectures and Protocols”, Prentice Hall.

24CS345T					Wireless Sensor Networks					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

PREREQUISITE: Computer Networks

COURSE OBJECTIVES

- To understand the concepts of sensor networks.
- To study the various protocols at various layers and its differences with traditional protocols.
- To understand the applications and security of Ad-Hoc and sensor networks.

UNIT 1: INTRODUCTION Key definitions of sensor networks, Advantages of sensor networks, Unique constraints and challenges, Driving applications, Enabling technologies for wireless sensor networks, Single-Node architecture, Hardware components, Energy consumption of sensor nodes, Network architecture -sensor network scenarios, Optimization goals and Figures of merit.	**10 Hrs.**
UNIT 2: MAC PROTOCOLS Introduction, Classifications of MAC protocols, Contention-based protocols, Contention-based protocols with reservation mechanisms, Contention-based MAC protocols with scheduling mechanisms, Other MAC protocols.	**11 Hrs.**
UNIT 3: ROUTING PROTOCOLS Introduction, Classification of routing protocols, Table-driven routing protocols, On- demand routing protocols, Hybrid routing protocols, Routing protocols with efficient flooding mechanisms, Hierarchical routing protocols, Power-aware routing protocols, Proactive routing.	**11 Hrs.**
UNIT 4: TRANSPORT LAYER AND SECURITY CHALLENGES Introduction to transport layer protocols, Classification of transport layer solutions, TCP over Ad Hoc wireless networks, Other transport layer protocol for Ad Hoc wireless networks. Security in Ad Hoc wireless networks, Network security requirements, Issues and challenges in security provisioning, Network security attacks, Secure routing in Ad Hoc wireless networks.	**10 Hrs.**
	42 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand the basics of wireless sensor networks and its applications.
CO2 - Understand typical node and network architectures.
CO3 - Compare different MAC layer protocols of the wireless sensor networks.
CO4 - Analyse routing layer protocols of the wireless sensor networks.
CO5 - Examine transport layer protocols of the wireless sensor networks.
CO6 - Identify and address the security issues in wireless sensor networks.

TEXT/REFERENCE BOOKS

1. C. Siva Ram Murthy and B.S.Manoj. “Ad-Hoc Wireless Networks: Architectures and Protocols”, Pearson Education.
2. Jagannathan Sarangapani. “Wireless Ad-hoc and Sensor Networks: Protocols, Performance and Control”, CRC Press.
3. Holger Karl & Andreas Willig, “Protocols and Architectures for Wireless Sensor Networks”, John Wiley.
4. Kazem Sohraby, Daniel Minoli, & Taieb Znati, “Wireless Sensor Networks- Technology, Protocols, and Applications”, John Wiley.